UP Special Topic 1 — Geography, Location, Boundaries, Physical Features, Climate, Soil and Drainage System

Current Affairs

Year	Fact	Why it matters
Rolling	Refresh edition-bound figures from latest Census / Survey / State CA desks.	Do not invent interim totals.

Teaching cards

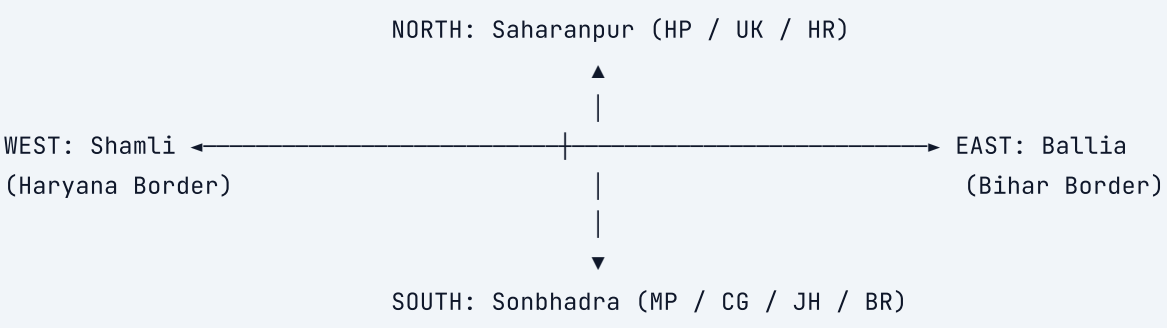
2.1 Geographical Location, Coordinates, and Regional Extremes

A. Absolute Location & Territorial Geometry

- **Latitudinal Coordinates**: Extends from **23^\circ 52′ ext{ N}** (southern tip of Sonbhadra) to **31^\circ 28′ ext{ N}** (northern boundary of Saharanpur). Latitudinal span = **7^\circ 36′**.
- **Longitudinal Coordinates**: Extends from **77^\circ 03′ ext{ E}** (western edge of Shamli) to **84^\circ 39′ ext{ E}** (eastern edge of Ballia). Longitudinal span = **7^\circ 36′**.
- **Geographical Dimensions**:
- East-to-West Length: **650 km**.
- North-to-South Width: **240 km**.
- **Total Surface Area**: **240,928 square kilometers** (constituting **7.33% of the total geographical area of India**).
- **National Area Rank**: Uttar Pradesh is the **4th largest State** in India by area (after Rajasthan, Madhya Pradesh, and Maharashtra).

B. The Four Geographic Extremities of Uttar Pradesh

- **Northernmost District**: **Saharanpur** (borders HP, Uttarakhand, Haryana).
- **Southernmost District**: **Sonbhadra** (borders MP, Chhattisgarh, Jharkhand, Bihar).
- **Easternmost District**: **Ballia** (surrounded on 3 sides by Bihar).
- **Westernmost District**: **Shamli** (borders Haryana).



C. District Area Extremes

- **Largest District by Area**: **Lakhimpur Kheri (7,680 km²)**, followed by Sonbhadra (6,905 km²), Hardoi (5,986 km²), and Sitapur (5,743 km²).
- **Smallest District by Area**: **Hapur (660 km²)**, followed by Bhadohi / Sant Ravidas Nagar (1,015 km²), Shamli (1,067 km²), and Ghaziabad (1,179 km²).

2.2 Climate, Rainfall Distribution, and Seasons in Uttar Pradesh

A. Climatic Classification

- Uttar Pradesh predominantly falls under the **Sub-tropical Humid Monsoon Climate** (*Cwg* under the Köppen climate classification, signifying mild dry winters and hot wet monsoons).
- **Three Distinct Seasons**: 1. **Summer Season (March to mid-June)**:
  - Extremely hot and dry; characterized by strong, scorching westerly winds called **"Loo"**.
  - Average maximum temperatures hover around 36^\circ ext{C} - 39^\circ ext{C}; peaks above **48^\circ ext{C} in Bundelkhand and Agra/Mathura**.
  - Highest recorded summer temperatures: **Agra and Jhansi**.
  - Lowest summer temperatures: **Bareilly**. 2. **Rainy / Monsoon Season (mid-June to September)**:
  - Receives approx. **85%–90% of its total annual precipitation** during this season via the **South-West Monsoon (Bay of Bengal branch)**, known locally as **"Purva"** (easterlies).
  - Rainfall distribution decreases systematically along two spatial gradients:
  - **East to West** (Eastern UP receives ~112 cm; Western UP receives ~65 cm).
  - **North to South** (Terai foothill belt receives ~150 cm; Bundelkhand receives ~80 cm).
  - **Highest Rainfall District**: **Gorakhpur** (average **184.7 cm**).
  - **Lowest Rainfall District**: **Mathura** (average **54.4 cm**). 3. **Winter Season (October to February)**:
  - Dry, sunny days and cold nights. January is the coldest month.
  - Occasional winter rainfall (10–15 cm) is caused by **Western Disturbances** originating in the Mediterranean Sea, highly beneficial for Rabi crops (Wheat, Mustard, Gram).

2.3 Major Lakes, Wetlands, and Natural Water Reservoirs of UP

Lake / Wetland / Water Body	District Located	Ecological Status / Special Significance
Gomat Taal (Fulhar Jheel)	**Pilibhit**	Origin source of the holy **Gomti River**.
Keetham Lake (Sur Sarovar)	**Agra**	Declared a **Ramsar Wetland Site (2020)**; wildlife sanctuary on Delhi-Agra highway.
Bakhira Wildlife Sanctuary & Lake	**Sant Kabir Nagar**	Declared a **Ramsar Wetland Site (2021)**; large freshwater natural lake.
Suraha Taal (Loknayak Bird Sanctuary)	**Ballia**	Massive crescent-shaped oxbow lake on the Ganga floodplain.

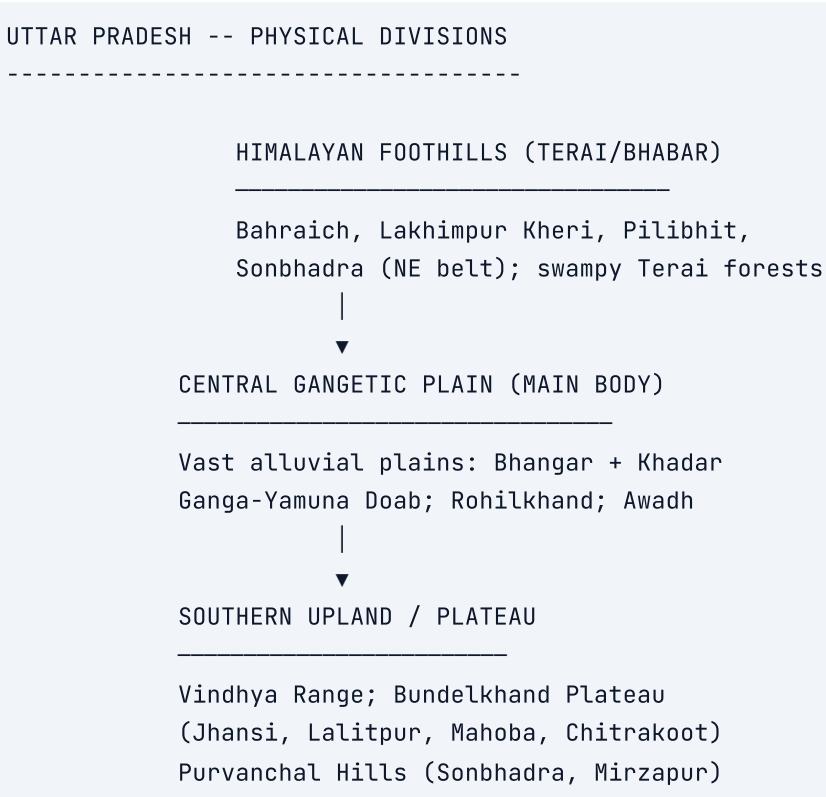
Lake / Wetland / Water Body	District Located	Ecological Status / Special Significance
Govind Ballabh Pant Sagar	**Sonbhadra** (on Rihand River)	**Largest artificial man-made freshwater reservoir in India** (Rihand Dam).
Samaspur Bird Sanctuary Lake	**Rae Bareli**	Ramsar Site; perennial freshwater wetland on Indo-Gangetic plain.
Sandi Bird Sanctuary Lake	**Hardoi**	Ramsar Site; natural freshwater wetland fed by the Garra river.
Parvati Arga Wetland	**Gonda**	Ramsar Site; twin oxbow lakes of the Saryu river.
Nawabganj Bird Sanctuary (Pt. Deendayal)	**Unnao**	Ramsar Site (2019); premier migratory water-bird sanctuary.
Sarsai Nawar Wetland	**Etawah**	Ramsar Site; permanent marshland hosting largest breeding population of Sarus Cranes.
Haiderpur Wetland	**Muzaffarnagar & Bijnor**	Declared **47th Ramsar Site of India (2021)**; human-made wetland on Madhya Ganga Canal.
Ramgarh Taal	**Gorakhpur**	Historic natural freshwater lake spanning ~1,700 acres; eco-tourism hub.
Radha Kund & Shyam Kund	**Mathura** (Govardhan)	Sacred pilgrimage lakes associated with Sri Krishna heritage.
Moti Jheel	**Kanpur**	Historic urban freshwater reservoir and drinking water source.
Raja Ka Bandh / Sultanpur Lake	**Sultanpur**	Natural floodplain reservoir.
Chittora Jheel	**Bahraich**	Historic lake associated with Raja Suheldev and Sai River tributary.

Bilingual Terminology -- UP Geography

English	Hindi	Key Anchor
Terai	तराई	Swampy foothills belt along Nepal border (Bahraich, Lakhimpur, Kheri)
Bhangar	बांगर	Old alluvial soil above flood level; firm, slightly elevated
Khadar	खादर	New alluvial soil deposited by annual floods; highly fertile
Usar	ऊसर / रेह	Saline-alkaline waste land common in central/eastern UP
Vindhya Plateau	विंध्य पठार	Southern UP; hard rock; limited agriculture
Bundelkhand Plateau	बुंदेलखंड पठार	SW UP; granite-gneiss rock; poor water retention
Purvanchal Hills	पूर्वांचल पहाड़ियाँ	Eastern edge of UP (Sonbhadra, Mirzapur)

English	Hindi	Key Anchor
Ganga-Yamuna Doab	गंगा-यमुना दोआब	Fertile plain between Ganga and Yamuna rivers
Rohilkhand	रुहेलखंड	Region: Bareilly, Moradabad, Pilibhit, Rampur, Budaun
Awadh	अवध	Central UP plains: Lucknow, Faizabad, Sultanpur
Braj	ब्रज	Mathura-Agra cultural heartland
Brij Bhoomi	ब्रजभूमि	Krishna's homeland; Mathura, Vrindavan, Govardhan
Ken River	केन नदी	Vindhyan river; joins Yamuna; Panna Tiger Reserve on banks
Betwa River	बेतवा नदी	Rises in MP; joins Yamuna at Hamirpur; Rajghat Dam
Sharda River	शारदा नदी	Also called Kali or Mahakali; flows from Nepal through Pilibhit

Extended Theory -- Physical Geography of UP



River Systems of UP

River	Origin	Joins	Key Districts / Dams
Ganga	Gangotri Glacier	Bay of Bengal	Haridwar--Varanasi corridor; Narora Barrage
Yamuna	Yamunotri Glacier	Ganga at Prayagraj (Triveni Sangam)	Mathura, Agra, Etawah

River	Origin	Joins	Key Districts / Dams
Gomti	Gomat Taal (Pilibhit)	Ganga at Ghazipur	Lucknow, Jaunpur, Ghazipur
Ghaghra / Saryu	Nepal Himalayas (Karnali)	Ganga at Chapra (Bihar)	Bahraich, Faizabad, Gonda
Gandak	Nepal	Ganga at Patna	NE border UP-Bihar
Betwa	Madhya Pradesh (Vindhyas)	Yamuna at Hamirpur	Jhansi region; Rajghat Dam
Ken	MP Vindhyas	Yamuna at Banda	Bundelkhand; Ken-Betwa river link project
Son	Amarkantak (MP)	Ganga at Patna	Sonbhadra; Obra Dam; Rihand Reservoir
Rihand	MP/Chhattisgarh	Son River	Rihand Dam / Govind Ballabh Pant Sagar
Sharda (Kali)	Nepal/Uttarakhand	Ghaghra at Bahramghat	Pilibhit, Lakhimpur, Sharda Sagar Dam

One-Liner Revision -- UP Geography

#	Fast Fact
1	UP = 9th largest State by area (**2,40,928 km2**)
2	UP shares borders with **9 States/UTs** + Nepal
3	Longest river in UP = **Ganga**
4	Gomti originates at **Gomat Taal (Pilibhit)**
5	Triveni Sangam (Ganga + Yamuna + Saraswati) = **Prayagraj**
6	Ken-Betwa river link project = Bundelkhand water transfer
7	Rihand Dam reservoir = **Govind Ballabh Pant Sagar** (largest by area)
8	Bhangar = **old alluvium** (above flood plain; firm soil)
9	Khadar = **new alluvium** (floodplain; very fertile)
10	Terai = swampy foothills belt; rich forests; wildlife
11	Usar = saline/alkaline wasteland (common in Etah, Mainpuri)
12	Vindhya plateau = southern UP (hard rock)
13	Bundelkhand = SW UP (granite-gneiss; water scarce)
14	Rohilkhand = Bareilly, Moradabad, Pilibhit, Rampur division

#	Fast Fact
15	Awadh = Lucknow, Faizabad, Sultanpur region